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# Design optimization of a multi-source renewable energy system using a novel method based on selective ensemble learning

Anas Bouaouda[a,*], Yassine Sayouti[a]

[a] LSIB Laboratory, FST Mohammedia, Hassan II University of Casablanca, Morocco

**Abstract**

This study conducts a comparative assessment to optimize the design of a Hybrid Renewable Energy System (HRES) consisting of PV panels, wind turbines (WTs), and hydrogen storage (PV/WT/FC). The study focuses on Dakhla City, leveraging its favorable weather conditions to evaluate the potential of renewable energy sources. A novel approach, Selective Ensemble Marine Predators (SEMPA), is introduced. SEMPA utilizes a selective ensemble learning strategy to enhance the performance of the Marine Predators Algorithm (MPA) and determine the optimal system size with the lowest Total Net Present Cost (TNPC) and improved reliability. To benchmark SEMPA's performance, we compare it against established methods, including MPA, Particle Swarm Optimization (PSO), and Harmony Search (HS). Results demonstrate that the integrated PV/WT/FC system provides a reliable and cost-effective energy solution for the southwest region of Morocco, particularly in Dakhla. SEMPA outperforms HS, PSO, and traditional MPA, achieving stable convergence after 30 iterations. The system exhibits a superior Cost of Energy (COE) at 0.3127 \$/kWh, a TNPC of 0.9689 M\$, and a low Loss of Power Supply Probability (LPSP) at 0.04747.



*Keywords:* Marine predators algorithm; Microgrid; Hybrid system; Metaheuristic; Design; Renewable energy

## 1. Introduction

For decades, the global energy landscape has been predominantly shaped by the extensive use of fossil fuels such as oil and coal. However, the associated drawbacks, including concerns about global warming, environmental pollution, and the volatility of fossil fuel prices, have prompted a significant shift in global perspective in recent years [1]. This shift, coupled with growing considerations for energy security, has led many nations to explore alternative energy sources. The objective is to establish cleaner, more cost-effective, and more reliable energy systems, marking a crucial transition away from the traditional reliance on fossil fuels [2].

Renewable energy resources, abundant and environmentally friendly, hold significant promise in offering affordable, sustainable, and clean energy solutions to combat the challenges of energy scarcity and climate change. Despite

* Corresponding author.
*E-mail address:* anas.bouaouda-etu@etu.univh2c.ma


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these advantages, the intermittent nature of individual renewable sources can pose reliability issues in standalone energy systems [3]. Addressing this challenge, the concept of Hybrid Renewable Energy Systems (HRES) emerges as a viable solution, particularly in remote areas where grid connections are often impractical, such as on remote islands. HRES integrates multiple energy sources, primarily renewable, and sometimes in conjunction with a grid, to meet the electricity demands of a region. By harnessing a variety of renewable resources, HRES mitigates the unreliability associated with a single energy source while maintaining low greenhouse gas emissions. This makes it an attractive option for sustainable energy generation [4].

A plethora of studies have delved into the design of HRESs, each employing unique approaches and optimization techniques [5]. Singh et al. [6] focused on optimizing the capacity of a hybrid system comprising an FC, electrolyzer, and PV, utilizing Artificial Bee Colony (ABC), Particle Swarm Optimization (PSO), and a hybrid of the two algorithms. Bouaouda et al. [7] proposed the Dandelion Optimizer to determine the optimal sizing of an off-grid PV/WT/FC system in a rural Moroccan region, incorporating a hydrogen tank as a backup energy source. Samy et al. [8] employed Harmony Search, inspired by the FireFly optimization algorithm, to optimize the design of a PV/WT/FC system catering to the energy demands of a tourist resort in Egypt. Gökçek et al. [9], on the other hand, conducted a techno-economic analysis of a hydrogen refueling station powered by solar and wind energy in Turkey, utilizing the HOMER software. These diverse studies exemplify the wide array of methodologies employed to address the complexities of hybrid renewable system design, tailored to different geographical contexts and energy requirements.

An extensive review of the literature reveals a diverse range of techniques utilized for optimizing HRES through the application of metaheuristics. However, in line with the No-Free-Lunch theorems [10], which posit the absence of a universally superior optimization method for all problems, the optimization of varied HRES configurations requires ongoing refinement and exploration of novel techniques. This theorem suggests that existing approaches can be improved, and innovative methods may provide valuable contributions. Therefore, the goal is to strive for the development of an efficient technique to address the HRES design problem, recognizing the dynamic nature of optimization challenges and the need for continuous methodological advancements.

This paper addresses the optimal sizing of a stand-alone HRES for an isolated region in Dakhla City, Southwest Morocco. The system utilizes renewable energy sources, including PV modules and WTs, integrated with a hydrogen system comprising fuel cells, electrolyzers, and a storage tank. Sizing optimization is achieved through an enhanced variant of the marine predator algorithm. The paper delves into the system configuration, unit-sizing considerations, and the computation of the Total Net Present Cost (TNPC) for the entire system. Additionally, simulation studies have been conducted to validate the performance of the designed system, comparing it against other well-established optimization algorithms commonly employed in the domain of HRES design problems.

The primary contributions of this paper can be summarized as follows:

- An improved version of the marine predator algorithm, called SEMPA, is proposed to optimize the sizing of an HRES in a rural area in Dakhla, Morocco.
- The proposed HRES, comprising PV panels, WTs, and a hydrogen storage system, is mathematically modeled to represent its complex dynamics.
- The optimization problem is formulated to minimize the TNPC while satisfying hybrid reliability constraints based on LPSP.
- The effectiveness of SEMPA is evaluated by comparing its performance with established optimization algorithms, such as HS and PSO methods.

The paper is organized as follows: Section 2 offers a succinct overview of the proposed HRES, incorporating mathematical equations related to system components. In Section 3, an economic analysis of the system is presented, followed by the introduction of the proposed method in Section 4. Section 5 delves into the discussion of simulation results. Finally, Section 6 summarizes the study's conclusions, highlighting key findings, and suggesting potential avenues for future research.

## 2. The framework of the proposed HRES

In this section, we present a mathematical model of the proposed HRES, which comprises a PV array, wind turbines, fuel cells, a hydrogen tank, an electrolyzer, and an inverter as shown in Figure 1.

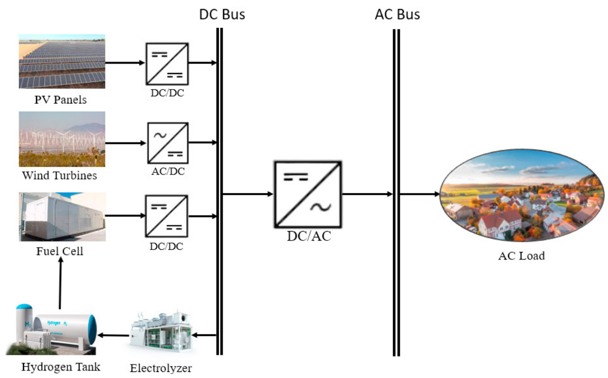



Fig. 1: Schematic diagram of the HRES

### 2.1. PV panel model

PV solar panels capture solar energy and convert it into electricity through the photovoltaic effect. Equation (1) expresses the PV system output power [7].

$$P_{PV} = N_{PV} \times P_r \times \eta_{PV} \times \left(\frac{G}{G_{ref}}\right) \times \left[1 + \beta\left(T - T_{ref}\right)\right] \tag{1}$$

### 2.2. WT model

The approximate estimation of the wind turbine's power output is calculated as follows [7]:

$$P_{WT}(t) = N_{WT} \times \begin{cases} 0, & \text{if } V(t) < V_{in} \text{ or } V(t) \geq V_{out} \\ V(t)^3 \left(\frac{P_r}{V_r^3 - V_{in}^3}\right) - P_r \left(\frac{V_{in}^3}{V_r^3 - V_{in}^3}\right), & \text{if } V_{in} \leq V(t) < V_r \\ P_r, & \text{if } V_r \leq V(t) < V_{out} \end{cases} \tag{2}$$

### 2.3. Electrolyzer

Powered by some of the electrical energy generated by WTs and PV panels, the electrolyzer harnesses water electrolysis to produce oxygen and hydrogen. This process entails passing an electric current through two submerged electrodes, effectively separating water into its fundamental components: oxygen and hydrogen. The electrolyzer's power output is represented in Equation (4) as follows [7]:

$$P_{el\_ht} = P_{ren\_el} \times \eta_{el} \tag{3}$$

### 2.4. Hydrogen storage tank model

The electrolyzer produces hydrogen, which is stored in a dedicated tank and subsequently fed to the fuel cell (FC) for electricity generation. The energy stored in the hydrogen at a given time $t$ can be estimated using Equation (4) [7].

$$E_{ht}(t) = E_{ht}(t-1) + P_{ren\_el}(t) \times \Delta t - P_{ht\_fc}(t) \times \Delta t \times \eta_{ht} \tag{4}$$

### 2.5. Fuel cell model

The FC acts as an energy storage system, bridging the gaps in power generation from WT and PV sources while ensuring a consistent supply to the load. Comprising a hydrogen storage tank and an electrolyzer, the FC facilitates chemical reactions that transform the stored hydrogen's chemical energy into electricity. This conversion involves a chemical reaction where the fuel oxidizes, generating electrical energy as a byproduct. The power output of the FC can be described as [7]:

$$P_{fc\_inv} = P_{ht\_fc} \times \eta_{fc} \tag{5}$$

## 3. Problem formulation

This section outlines the formulation of the objective function and the constraints for the design of the PV/WT/FC system.

### 3.1. Objective function

The main objective is to minimize the TNPC associated with the components within the PV/WT/FC system.

$$\begin{aligned} &ObjectiveFunction = Min.TNPC\left(N_{WT}, N_{PV}, N_{el}, N_{tank}, N_{fc}\right) = \\ &\sum_{i=WT,PV,el,tank,fc} TNPC_i = N_i \times \left( \begin{array}{l} C_{capital,i} + C_{replacement,i} \times \sum_{n=1}^{y_i} \frac{1}{(1+ir)^{n.L_i}} \\ +C_{maintenance,i} \times \frac{(1+ir)^R - 1}{ir(1+ir)^R} \end{array} \right) \end{aligned} \tag{6}$$

where $i$ refers to the specific equipment, $N$ signifies the number of units of that equipment, $C_{capital,i}$ denotes the investment cost, $C_{replacement,i}$ represents the replacement cost, and $C_{maintenance,i}$ stands for the annual maintenance and repair cost over the project's duration, indicated by $R$. Additionally, $ir$ denotes the real interest, calculated from the nominal interest $ir_{nominl}$ and the annual inflation rate $(f)$. Within the equation, $L$ and $y$ correspond to the equipment's useful lifespan and the number of replacements, respectively. Therefore, the objective in designing the hybrid system is to minimize TNPC while ensuring the reliability of LPSP.

### 3.2. Constraints

The optimization of the objective function is subject to the following constraints:

$$0 \leq LPSP \leq LPSP_{max} \tag{7}$$

$$0 \leq N_i \leq N_i^{max} \tag{8}$$

$$E_{\tan\ k}(0) \leq E_{\tan\ k}(8760) \tag{9}$$

Within the aforementioned constraints, $LPSP_{max}$ denotes the maximum value for the reliability of LPSP, and $N_i^{max}$ represents the maximum allowable quantity of equipment within the system.

## 4. Optimization Method

To address the formulated problem, a robust and reliable optimization algorithm is essential. The primary structure of the proposed SEMPA is outlined in the following subsections.

### 4.1. Overview of Marine Predators algorithm

Derived from the intricate movements of marine predators and their prey, the Marine Predators algorithm (MPA) is a metaheuristic optimization technique designed to emulate their adaptive foraging strategies [11]. This algorithm dynamically transitions between Lévy and Brownian motion based on the iteration interval, closely mirroring the behavior of predators. Inspired particularly by the foraging patterns of silky sharks around drifting Fish Aggregating Devices (FADs), MPA employs Brownian motion in prey-rich areas and regions with low hunting pressure. In zones characterized by moderate hunting conditions, predators in MPA utilize a combination of both Lévy and Brownian motion, showcasing their adaptability to varying environmental conditions [11]. The mathematical representation of the MPA method is as follows:

#### 4.1.1. Initialization

Similar to many metaheuristic algorithms, the MPA starts by initializing the solution uniformly across the search space [11].

$$X_{i,j} = X_{l,j} + r_1 \times \left(X_{u,j} - X_{l,j}\right), i = 1, 2, 3, \ldots, n. \quad j = 1, 2, ..Dim \tag{10}$$

#### 4.1.2. Phase 1

In the exploration phase, characterized by the optimal strategy for a predator to remain stationary, two equations are proposed to model movement, considering the inherent characteristic that predators move at a slower pace than their prey [11]:

While $Iter < \frac{1}{3} Max\_Iter$

$$S\vec{S}_i = \vec{R}_B \otimes \left(\vec{E}_i - \vec{R}_b \otimes Pr\vec{e}y_i\right); \qquad Pr\vec{e}y_i = Pr\vec{e}y_i + 0.5 * \vec{R} \otimes S\vec{S}_i \tag{11}$$

#### 4.1.3. Phase 2

This phase represents the gradual transition from exploration to exploitation. Consequently, it involves the integration of both Brownian and Levy distributions. Two subsets are generated from the original population, aligning with the updated population in both the exploration and exploitation phases [11].

While $\frac{1}{3} Max\_iter \leq Iter \leq \frac{2}{3} Max\_iter$, the equations for two groups are provided in Equations. (12) and (13) [11]. For the first group:

$$S\vec{S}_i = \vec{R}_L \otimes \left(\vec{E}_i - \vec{R}_L \otimes Pr\vec{e}y_i\right), i = 1.2. \ldots n/2; \qquad Pr\vec{e}y_i = Pr\vec{e}y_i + 0.5 * \vec{R} \otimes S\vec{S}_i \tag{12}$$

For the second group:

$$S\vec{S}_i = \vec{R}_B \otimes \left(\vec{R}_B \otimes \vec{E}_i - Pr\vec{e}y_i\right), i = n/2 + 1. \ldots n; \qquad Pr\vec{e}y_i = \vec{E}_i + 0.5 * CP \otimes S\vec{S}_i \tag{13}$$

#### 4.1.4. Phase 3

This phase corresponds to the exploitation stage, characterized by predators moving at a slower pace than their prey [11].

While $Iter > \frac{2}{3} Max\_iter$

$$S\vec{S}_i = \vec{R}_L \otimes \left(\vec{R}_L \otimes \vec{E}_i - Pr\vec{e}y_i\right); \qquad Pr\vec{e}y_i = \vec{E}_i + 0.5 * CP \otimes S\vec{S}_i \tag{14}$$

There are two additional key considerations in the MPA, namely:

### 4.2. Eddy formation and the effect of FADS:

Another crucial factor influencing marine predators' behavior is the effect of Fish Aggregating Devices (FADs). FADs are treated as local optima, and their impact is to guide the search towards these points in the search space. The representation of the FADs effect is outlined as follows [11]:

$$Pr\vec{e}y_i = \begin{cases} Pr\vec{e}y_i + CP * \left[\vec{X}_l + \vec{R} \otimes \left(\vec{X}_u - \vec{X}_l\right)\right] \otimes \vec{U} & , r \leq FADs \\ Pr\vec{e}y_i + [FADs * (1 - r) + r] * (Pr\vec{e}y_{r1} - Pr\vec{e}y_{r2}) & , r > FADs \end{cases} \tag{15}$$

### 4.3. Marine memory

Moreover, the MPA integrates predator memory features into the search process, allowing predators to recall their most successful foraging locations. This mechanism functions as a repository where optimal iterative solutions are stored and utilized for updating the Elite matrix. For a more in-depth understanding, readers are referred to Faramarzi et al. [11].

### 4.4. Proposed algorithm

The MPA algorithm has two significant weaknesses: it does not effectively explore all possible solutions within the search space, and it is prone to premature convergence because its optimization iterations are divided into three distinct parts: exploration, transition from exploration to exploitation, and exploitation. To address these issues, the Selective Ensemble Marine Predator Algorithm (SEMPA) was developed. SEMPA integrates a selective ensemble strategy into MPA to prevent premature convergence and increase exploration within the search space.

Many researchers integrate metaheuristic techniques with machine learning to address complex optimization problems, a strategy validated in numerous studies [12, 13, 14]. Ensemble learning, which combines multiple strategies with diverse characteristics to improve overall performance, is one of these approaches. The novel concept of selective ensemble, proposed in this context, involves judiciously selecting strategies to avoid wasting resources. In other words, not all strategies are used indiscriminately to optimize performance. This paper integrates the selective ensemble concept with the MPA. If a phase within MPA is deemed suitable for the ongoing search, its selection probability is increased. Conversely, if a strategy proves unsuitable, its selection probability is decreased. A priority roulette selection method is devised for phase selection, facilitating the adaptive adjustment of selection probabilities within the algorithm.

## 5. Results and Discussion

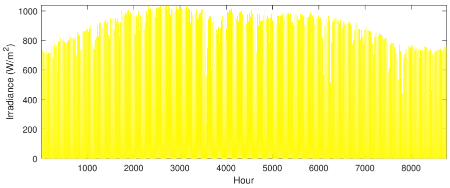
This study explores the integration of an HRES incorporating PV panels, WT, and a hydrogen energy storage system. The investigation focuses on the techno-economic analysis and optimization of a PV/WT/FC system in Dakhla, situated in the southern region of Morocco (latitude: 23.9337, longitude: -15.6694). The primary objective is to minimize the TNPC while ensuring reliability indices. Solar radiation, hourly temperature, hourly wind speed, and the hourly load profile (IEEE RTS load with a 50 kW peak) throughout the year are illustrated in Figures 2a, 2b, 2c, and 2d, respectively. Table 1 presents the economic details of the HRES components.

Table 1: Economic details regarding the components of the microgrid system [3, 7].

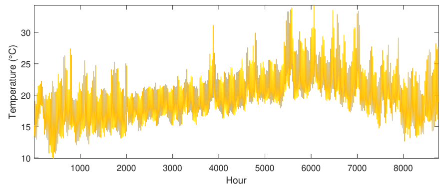
| Component | IC (US$/unit) | O&MC (US$/unit-yr) | RC (US$/unit) | Rated Capacity | Lifetime (Year) |
|---|---|---|---|---|---|
| Wind Turbine | 3200 | 32 | 3000 | 1 kW | 20 |
| PV panel | 2000 | 20 | 1800 | 1 kW | 20 |
| Inverter | 800 | 7 | 750 | 1 kW | 15 |
| Electrolyzer | 2000 | 20 | 1500 | 1 kW | 20 |
| Hydrogen tank | 1300 | 15 | 1200 | 1 Kg | 20 |
| Fuel cell | 3000 | 40 | 2500 | 1 kW | 5 |

The proposed methodology was implemented using MATLAB, with project parameters set as follows: a useful lifetime of 20 years, an interest rate of 6%, a nominal interest rate of 9%, and an inflation rate of 3%. To assess its effectiveness, SEMPA was compared with conventional optimization algorithms, including MPA, HS, and PSO, which are widely employed in various optimization problems. Each optimization method underwent 30 independent implementations, with a fixed population size and maximum iteration of 50 and 100, respectively, for all algorithms. The optimization variables encompassed the number of wind WTs, PV panels, inverter power for load transfer, fuel cell power, electrolyzer power, and hydrogen tank mass, all within a permissible range between zero and 2000.

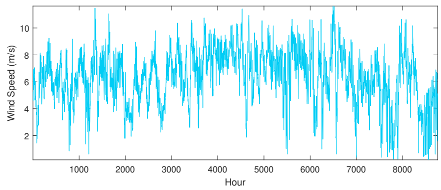
Figure 3 presents a comparison of the convergence curves for the best-performing execution among 30 runs of the optimization algorithms. Analyzing the convergence patterns, SEMPA demonstrates superior performance, achieving the minimum fitness value after only 30 iterations, outperforming MPA, HS, and PSO.



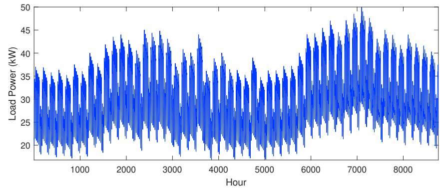

(a) Hourly irradiance




(b) Hourly temperature




(c) Hourly wind speed




(d) Hourly load demand

Fig. 2: Meteorological Data

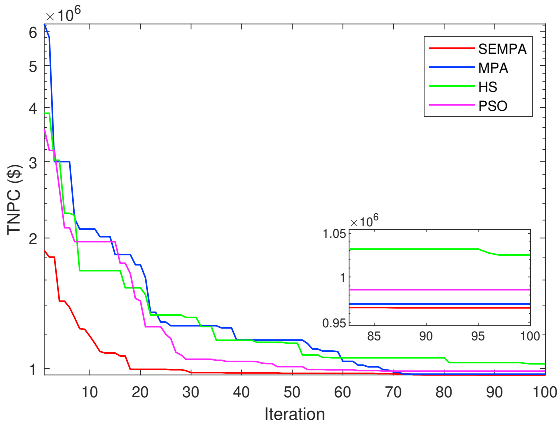



Fig. 3: Convergence curve of different algorithms

To comprehensively compare all methods, the results obtained from 30 independent runs for the considered case underwent rigorous statistical analyses, employing both parametric and non-parametric assessments. Parametric measures encompassed the best, worst, and mean values, while non-parametric evaluations incorporated standard deviation (Std) and ranking. The statistical outcomes for the SEMPA, MPA, HS, and PSO algorithms are summarized in Table 2. Significantly, the results highlight the superiority of the SEMPA algorithm across all metrics, surpassing the performance of HS, PSO, and the standard MPA.

Table 2: Statistical results

| System configuration | Index | Algorithm | | | |
|---|---|---|---|---|---|
| | | SEMPA | MPA | HS | PSO |
| PV/WT/FC | Best | **9.6275E+05** | 9.6886E+05 | 9.9043E+05 | 9.6634E+05 |
| | Worst | **9.7865E+05** | 1.0937E+06 | 1.2147E+06 | 1.1043E+06 |
| | Mean | **9.6799E+05** | 9.9447E+05 | 1.0664E+06 | 1.0102E+06 |
| | Std | **4.0225E+03** | 3.0977E+04 | 5.8510E+04 | 3.9850E+04 |
| | Rank | **1** | 2 | 3 | 4 |

The numerical results for the optimal design of the HRES with a $LPS\,P_{max} = 5\%$ are presented in Table 3. Analysis of the results reveals that the proposed SEMPA achieves a minimum TNPC of 0.9689 M$ and a COE of 0.3127 $/kWh.

These values adhere to the predefined LPSP constraint ($LPSP \leq LPSP_{max}$), with a calculated value of 0.04747. The outcomes underscore SEMPA's superior performance in achieving a more reliable and cost-effective design for the PV/WT/FC system compared to the MPA, PSO, and HS algorithms, demonstrating higher reliability and lower cost.

Table 3: Results of optimal design of the HRES obtained using various methods

| System | Algorithm | $N_{pv}$ | $N_{wt}$ | $P_{el}$ (kW) | $P_{fc}$ (kW) | $M_{HTS}$ (kg) | $P_{inv}$ (kW) | TNPC (M$) | LPSP | CEO ($/kWh) |
|---|---|---|---|---|---|---|---|---|---|---|
| PV/WT/FC | SEMPA | **47** | **55** | **35.39** | **27.90** | **84.35** | **48.11** | **0.9627** | **0.04647** | **0.3115** |
| | MPA | 42 | 57 | 40.39 | 26.84 | 92.74 | 48.65 | 0.9689 | 0.04747 | 0.3127 |
| | HS | 47 | 55 | 28.17 | 30.96 | 84.48 | 48.77 | 0.9904 | 0.04759 | 0.3215 |
| | PSO | 54 | 52 | 36.82 | 28.20 | 79.73 | 48.25 | 0.9663 | 0.04861 | 0.3143 |

## 6. Conclusion

This study introduces a modified version of the Marine Predators Algorithm, termed Selective Ensemble Marine Predators (SEMPA), tailored for optimizing a PV/WT/FC system. Addressing the challenge of power supply failure probability, SEMPA aims to determine the optimal capacity of renewable resources while minimizing the TNPC. Comparative evaluations, covering generating cost, renewable sources, and convergence time, reveal that the proposed optimization approach surpasses conventional MPA, HS, and PSO methods. The findings unequivocally establish the efficacy of the proposed method in reducing costs and enhancing reliability. Future research directions could involve refining the estimation of renewable energy supply and demand, particularly in light of limited historical data. Additionally, optimizing computational efficiency through advanced variance reduction techniques, such as common random numbers, could be explored.

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